

Lab 09

Object Oriented Programming

BS DS Fall 2025 Morning

Instructions:

- Indent your code properly.
- Use meaningful variable names and follow standard naming conventions.
- Use clear and meaningful prompts/labels for all program input and output.
- Make sure that there are NO dangling pointers or memory leaks in your program.
- Do not discuss your problems with fellow students. If you face any difficulty in understanding, consult your TAs.

Task 1: Parking Ticket Simulator

For this assignment you will design a set of classes that work together to simulate a

police officer issuing a parking ticket. The classes you should design are:

• **The ParkedCar Class:** This class should simulate a parked car. The class's responsibilities are:

- To know the car's make, model, color, license number, and the number of minutes that the car has been parked

• **The ParkingMeter Class:** This class should simulate a parking meter. The class's only responsibility is:

- To know the number of minutes of parking time that has been purchased

• **The ParkingTicket Class:** This class should simulate a parking ticket. The class's responsibilities are:

- To report the make, model, color, and license number of the illegally parked car
- To report the amount of the fine, which is \$25 for the first hour or part of an hour that the car is illegally parked, plus \$10 for every additional hour or part of an hour that the car is illegally parked
- To report the name and badge number of the police officer issuing the ticket

• **The PoliceOfficer Class:** This class should simulate a police officer inspecting parked cars. The class's responsibilities are:

- To know the police officer's name and badge number
- To examine a ParkedCar object and a ParkingMeter object, and determine whether the car's time has expired
- To issue a parking ticket (generate a ParkingTicket object) if the car's time has expired

Write a program that demonstrates how these classes collaborate.

Task 2: Carpet Calculator

The Nayyer Carpet Company has asked you to write an application that calculates the price of carpeting for rectangular rooms. To calculate the price:

- Multiply the area of the floor (width times length) by the price per square foot of carpet.

For example, the area of a floor that is 12 feet long and 10 feet wide is 120 square feet. To cover that floor with carpet that costs Rs. 80 per square foot would cost Rs. 9600 ($12 * 10 * 80 = 9600$).

Note: You will need the **FeetInches** class provided with this Lab in [FeetInches.cpp](#) file.

RoomDimension Class

Create a class named **RoomDimension** with the following functionalities:

1. Two private data members:

- A **FeetInches** object named length that holds the length of the room in feet.
- A **FeetInches** object named width that holds the width of the room in feet.

2. Provide implementation for the following functions:

- Mutators for all data members (length and width) of the class.
- Accessors for all data members (length and width) of the class.
- Constructor accepting length and width as arguments.
- Member function that returns the area of the room in square feet.
- Member function that displays the length and width of the appropriate object.

RoomCarpet Class

Create a class named **RoomCarpet** with the following functionalities:

1. Two private data members:

- A **RoomDimension** object named roomSize that holds the dimensions of the room.
- A float named **costPerSquareFoot** that holds the cost of the carpet per square foot.

2. Provide implementation for the following functions:

- Mutators for all data members (roomSize and costPerSquareFoot) of the class.
- Accessors for all data members (roomSize and costPerSquareFoot) of the class.
- Constructor accepting roomSize and costPerSquareFoot as arguments.
- Member function that returns the total cost of the carpet.
- Member function that displays the roomSize and costPerSquareFoot of the appropriate object.

Driver Program

Once you have written these classes, use them in an application that asks the user to enter the dimensions of a room and the price per square foot of the desired carpeting. The application should display the total cost of the carpet.